
Section D

You are advised to spend no more than **70 minutes** on this section.

Type your answers to **Section D** in your Electronic Answer Document.
You **must save** this document at regular intervals.

These questions require you to load the **Skeleton Program** and to make programming changes to it.

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This question refers to the subroutine `InputCoordinate` in the `Simulation` class.

The warren and fox inspection options in the **Skeleton Program** do not currently check if the coordinates entered by the user are on the landscape. This behaviour needs to be improved so that an error message is displayed if the user inputs coordinates for a location that is not on the landscape.

If the user runs a simulation with default settings then the landscape size is 15, so valid locations have an x coordinate between 0 and 14, inclusive.

What you need to do

Modify the `InputCoordinate` subroutine in the `Simulation` class so that, if a coordinate outside the range defined by the landscape size is input, the error message "Coordinate is outside of landscape, please try again." is displayed and the user is forced to re-input the coordinate.

To achieve full marks for this question, the `InputCoordinate` subroutine should work correctly for any landscape size, not just the default size of 15.

Test

Test your changes work by running the **Skeleton Program** and selecting the following options:

- "1. Run simulation with default settings"
- "3. Inspect fox"

Then input these three x coordinates for the location of the fox to inspect:

- -1
- 15
- 0

Evidence that you need to provide

Include the following in your Electronic Answer Document.

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Your PROGRAM SOURCE CODE for the amended subroutine
InputCoordinate.

[4 marks]

0	9	.	2
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SCREEN CAPTURE(S) for the described test.

Ensure that in your SCREEN CAPTURE(S) it can be seen that the x coordinates
-1 and 15 are rejected **and** that the x coordinate 0 is accepted, **and** that after
the 0 is input the **Skeleton Program** advances to ask the user to input the y
coordinate.

[1 mark]

Turn over for the next question

Turn over ►